DEPARTMENT OF COMPUTER SCIENCE

Subject: Database Systems

PROJECT REPORT

Structured Genomic Surveillance System for Rare Disorders (SGSS-RD)

Topic: Somatic Mosaicism — Tracking and Genomic Surveillance

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SGSS-RD | Structured Genomic Surveillance System for Rare Disorders

1. Introduction

1.1 Background:

Genetic diseases caused by somatic mosaicism represent one of the most underdiagnosed categories in modern medicine. Somatic mosaicism occurs when a post-zygotic DNA mutation arises after fertilisation, resulting in two or more genetically distinct cell populations coexisting within a single individual. Unlike germline mutations which are present in every cell from conception, mosaic mutations affect only a subset of cells, distributed according to when and where in embryonic development the error occurred.Standard diagnostic pipelines rely on blood-derived DNA which reliably captures germline variants but frequently misses tissue-specific mosaic mutations present at low variant allele fractions.

1.2 What is Somatic Mosaicism?

Every human begins as a single fertilized cell containing a complete diploid genome. As that cell divides and differentiates into the trillions of cells forming a complete organism, DNA replication errors occasionally arise. If such an error occurs in a progenitor cell early in embryogenesis, the mutation will be inherited by all descendant cells of that lineage. The result is a mosaic organism: genetically normal cells and mutant cells coexist in proportions determined by the timing and location of the original copying error.

1.3 Project Overview:

The Structured Genomic Surveillance System for Rare Disorders (SGSS-RD) is a relational database designed specifically to address the diagnostic and monitoring challenges of somatic mosaicism. It stores patient profiles, tissue sample metadata, mosaic variant cells with VAF measurements, formal rare-disease diagnoses, and longitudinal surveillance logs. By recording VAF across multiple tissue types and time points, SGSS-RD enables clinicians and researchers to detect mutation spread, trigger alerts when VAF crosses clinical thresholds, and build evidence bases for conditions that are individually rare but collectively significant.

1.4 Objectives:

- **Store multi-tissue variant data:** Record sequencing results and mosaic variants identified from different organ biopsies of the same patient.
- **Track VAF over time:** Log Variant Allele Frequency at each clinical visit and compute change relative to the previous measurement.
- **Measure diagnostic delay:** Record and analyse the number of years patients waited for a correct diagnosis to identify systemic healthcare gaps.
- **Support research queries:** Enable aggregate queries across patients, genes, organs, and countries for epidemiological and translational research.

1.5 Scope:

SGSS-RD is designed for academic and research use within a bioinformatics curriculum context. The schema is implemented in MySQL and models all core entities required for somatic mosaicism surveillance. Patient names are stored as plain text for demonstration; a clinical deployment would replace identifiable fields with anonymisation tokens and implement role-based access control and regulatory compliance infrastructure.

2. System Design and Architecture

2.1 Design Philosophy:

SGSS-RD is built on three core principles. Normalization eliminates redundancy by decomposing data into entities at appropriate granularity with foreign key relationships. Surveillance-first architecture gives the schema a dedicated longitudinal table that stores VAF at each clinical visit, the computed change from the previous measurement, and an automated alert flag, making trend detection a first-class database operation. Clinical standardization aligns attribute names and values with internationally recognized nomenclature to facilitate future integration with external resources.

2.2 Entity Identification:

Analysis of the somatic mosaicism domain identified five core entities, each with independent existence and multiple attributes of interest:

- **Patient:** The human individual receiving clinical care. The central entity around which all others are organized.
- **Tissue Sample:** A biological specimen collected from a specific organ at a specific time. The material substrate for sequencing.
- **Mosaic Variant:** A genetic mutation identified in a tissue sample, characterized by gene, mutation type, and critically its VAF percentage.
- **Disease Diagnosis:** A formal attribution of a named rare disease to a patient, including confirmation method and diagnostic delay.
- **Surveillance Log:** A time-stamped record of a VAF measurement at a follow-up visit, capturing change from the preceding measurement and any alert condition.

3. Database Schema

3.1 patients

The root entity of the schema. Every other table references it directly or indirectly. Patient identity is separated from clinical and genomic data to support de-identification workflows in production deployments.

Key	Attribute	Type	Constraint	Description
PK	patient_id	INT	AUTO_INCREMENT, NOT NULL	Unique numeric identifier for each patient
	patient_name	VARCHAR(50)	NOT NULL	Full name of the patient
	age	INT	NOT NULL	Age of the patient in years at time of registration
	gender	VARCHAR(10)	NOT NULL	Biological sex: Male, Female, or Other
	country	VARCHAR(50)	NOT NULL	Country of residence for epidemiological analysis

3.2 tissue_samples

Each row records one biological specimen. A patient may have samples from multiple organs at multiple time points. The sample_method affects the expected VAF range of any variants detected.

Key	Attribute	Type	Constraint	Description
PK	sample_id	INT	AUTO_INCREMENT, NOT NULL	Unique identifier for each biological sample
FK	patient_id	INT	NOT NULL, FK -> patients	Links sample to its source patient
	organ_name	VARCHAR(50)	NOT NULL	Organ sampled: Brain, Skin, Blood, Liver, Muscle, Spinal Cord
	sample_date	DATE	NOT NULL	Calendar date the sample was collected
	sample_method	VARCHAR(20)	NOT NULL	Collection technique: Biopsy, Blood draw, or Saliva

3.3 mosaic_variants

The central scientific table. The vaf_percent column is the most clinically significant attribute in the entire schema, quantifying what fraction of cells in the sampled tissue carry the mutation and defining the mosaic state.

Key	Attribute	Type	Constraint	Description
PK	variant_id	INT	AUTO_INCREMENT, NOT NULL	Unique identifier for each variant record
FK	sample_id	INT	NOT NULL, FK -> tissue_samples	Links variant to the sample in which it was detected
	gene_name	VARCHAR(30)	NOT NULL	HGNC gene symbol of the affected gene (e.g. MTOR, HRAS, GNAQ)
	mutation_type	VARCHAR(20)	NOT NULL	Category of DNA change: SNV, Indel, or CNV
	vaf_percent	DECIMAL(5,2)	NOT NULL	Variant Allele Frequency: percentage of reads carrying the mutation (0.00 to 100.00)
	is_harmful	VARCHAR(10)	NOT NULL	Pathogenicity assessment: Yes (pathogenic), No (benign), Unknown (VUS)

3.4 disease_diagnosis

Records the formal rare disease diagnosis issued to a patient. The years_to_diagnose attribute captures the diagnostic odyssey length — the years between symptom onset and confirmed diagnosis.

Key	Attribute	Type	Constraint	Description
PK	diagnosis_id	INT	AUTO_INCREMENT, NOT NULL	Unique identifier for each diagnosis record
FK	patient_id	INT	NOT NULL, FK -> patients	Links diagnosis to the patient
	disease_name	VARCHAR(100)	NOT NULL	Full clinical name of the rare disease (e.g. Focal Cortical Dysplasia, Sturge-Weber Syndrome)
	diagnosis_date	DATE	NOT NULL	Date on which the diagnosis was formally confirmed
	how_confirmed	VARCHAR(20)	NOT NULL	Confirmation method: DNA Test, Clinical exam, or Both
	years_to_diagnose	INT	NOT NULL	Years elapsed between first symptom presentation and confirmed diagnosis

3.5 surveillance_log

The surveillance_log implements the temporal monitoring capability that distinguishes SGSS-RD from a static variant registry. Every clinical follow-up visit produces a new row. The alert field is set to Yes when vaf_change exceeds 5 percentage points.

Key	Attribute	Type	Constraint	Description
PK	log_id	INT	AUTO_INCREMENT, NOT NULL	Unique identifier for each surveillance entry
FK	patient_id	INT	NOT NULL, FK -> patients	Links entry to the monitored patient
FK	variant_id	INT	NOT NULL, FK -> mosaic_variants	Links entry to the specific variant being tracked
	check_date	DATE	NOT NULL	Date of the clinical visit
	vaf_recorded	DECIMAL(5,2)	NOT NULL	VAF percentage measured at this visit
	vaf_change	DECIMAL(5,2)	NOT NULL	Since previous log entry; positive means mutation is spreading
	alert	VARCHAR(5)	NOT NULL	Yes if vaf exceeds 5.00%

4. Entity Relationship Diagram

4.1 ERD Overview:

The Entity Relationship Diagram for SGSS-RD illustrates five entity tables and five relationships connecting them. All relationships are one-to-many, reflecting the real-world cardinality of the domain. The ERD uses crow's-foot notation with a double vertical tick on the mandatory one-side and a circle-plus-crow's-foot on the optional many-side. All connector lines use straight horizontal and vertical routing to eliminate directional ambiguity. The full ERD is provided in the accompanying PDF diagram file.

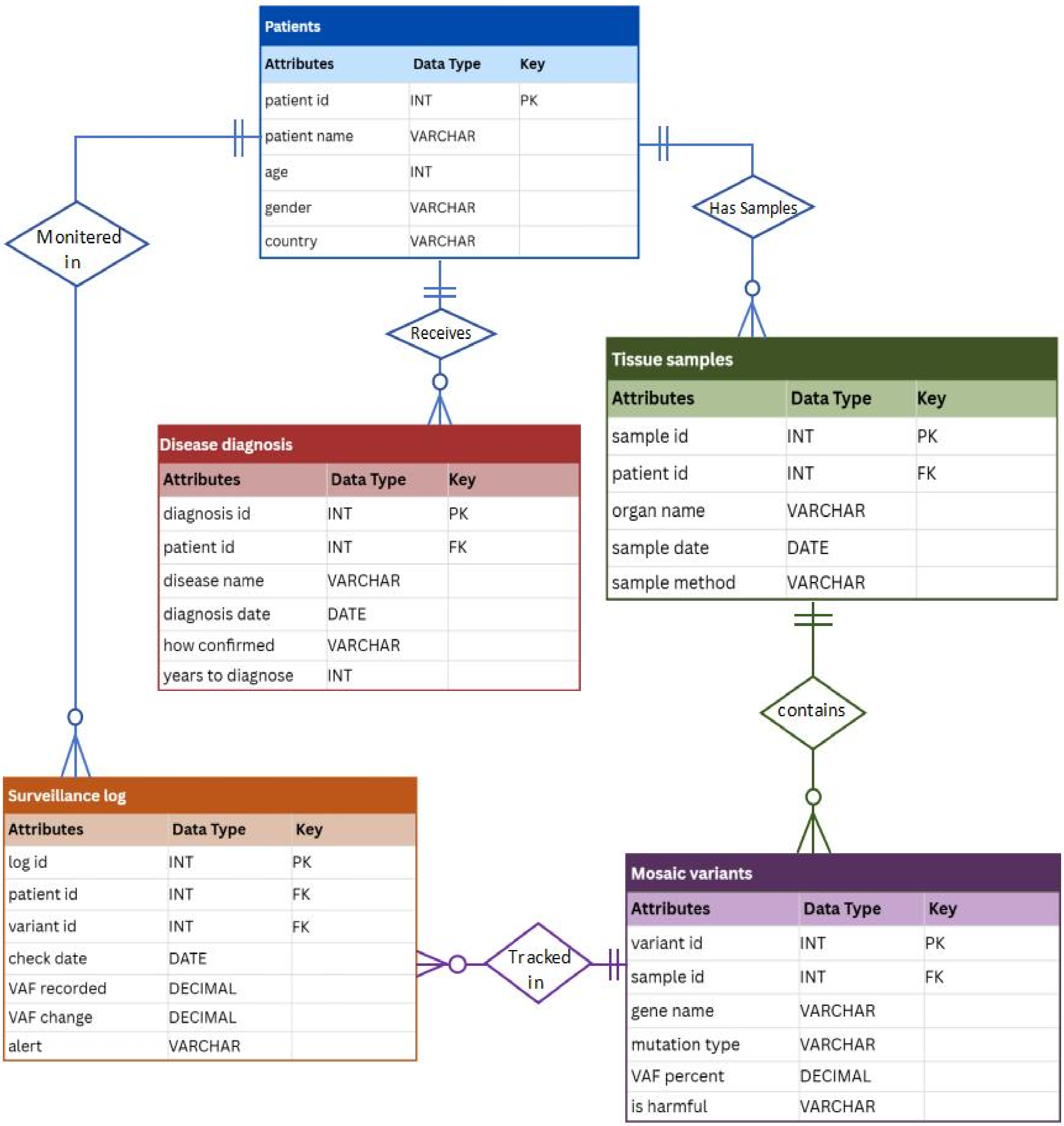
4.2 Cardinality Constraints:

Parent Table	Child Table	Cardinality	Business Rule
patients	tissue_samples	1 : Many (1:M)	One patient yields zero or more tissue samples
patients	disease_diagnosis	1 : Many (1:M)	One patient receives one or more diagnoses
patients	surveillance_log	1 : Many (1:M)	One patient has zero or more log entries
tissue_samples	mosaic_variants	1 : Many (1:M)	One sample contains zero or more variants
mosaic_variants	surveillance_log	1 : Many (1:M)	One variant generates zero or more log entries

4.3 Referential Integrity

All foreign key relationships are enforced at the database level using MySQL FOREIGN KEY constraints. This ensures that no tissue sample can be inserted without referencing a valid patient, no mosaic variant can reference a non-existent sample, no diagnosis can be assigned to an unregistered patient, and no surveillance log entry can reference a non-existent variant or patient. These constraints guarantee data consistency throughout the schema and prevent orphaned records.

Entity Relationship Diagram:



5. SQL Implementation

5.1 Table Creation

```sql
CREATE DATABASE sgss_rd;
USE sgss_rd;

CREATE TABLE patients (
    patient_id    INT PRIMARY KEY AUTO_INCREMENT,
    patient_name  VARCHAR(50)  NOT NULL,
    age           INT          NOT NULL,
    gender        VARCHAR(10)  NOT NULL,
    country       VARCHAR(50)  NOT NULL
);

CREATE TABLE tissue_samples (
    sample_id      INT PRIMARY KEY AUTO_INCREMENT,
    patient_id     INT         NOT NULL,
    organ_name     VARCHAR(50) NOT NULL,
    sample_date    DATE        NOT NULL,
    sample_method  VARCHAR(20) NOT NULL,
    FOREIGN KEY (patient_id) REFERENCES patients(patient_id)
);

CREATE TABLE mosaic_variants (
    variant_id     INT PRIMARY KEY AUTO_INCREMENT,
    sample_id      INT          NOT NULL,
    gene_name      VARCHAR(30)  NOT NULL,
    mutation_type  VARCHAR(20)  NOT NULL,
    vaf_percent    DECIMAL(5,2) NOT NULL,
    is_harmful     VARCHAR(10)  NOT NULL,
    FOREIGN KEY (sample_id) REFERENCES tissue_samples(sample_id)
);

CREATE TABLE disease_diagnosis (
    diagnosis_id      INT PRIMARY KEY AUTO_INCREMENT,
    patient_id        INT          NOT NULL,
    disease_name      VARCHAR(100) NOT NULL,
    diagnosis_date    DATE         NOT NULL,
    how_confirmed     VARCHAR(20)  NOT NULL,
    years_to_diagnose INT          NOT NULL,
    FOREIGN KEY (patient_id) REFERENCES patients(patient_id)
);

CREATE TABLE surveillance_log (
    log_id       INT PRIMARY KEY AUTO_INCREMENT,
    patient_id   INT          NOT NULL,
    variant_id   INT          NOT NULL,
    check_date   DATE         NOT NULL,
    vaf_recorded DECIMAL(5,2) NOT NULL,
    vaf_change   DECIMAL(5,2) NOT NULL,
    alert        VARCHAR(5)   NOT NULL,
    FOREIGN KEY (patient_id) REFERENCES patients(patient_id),
    FOREIGN KEY (variant_id) REFERENCES mosaic_variants(variant_id)
);
```

5.2 Sample Data Insertion

Fifty rows of representative data were inserted across all five tables. A representative subset is shown below; the complete data set is provided in the accompanying SQL file SGSS_RD_database.sql.

```
-- patients
INSERT INTO patients (patient_name, age, gender, country) VALUES
('Ahmed Raza', 7, 'Male', 'Pakistan'),
('Sara Khan', 34, 'Female', 'Pakistan'),
('John Miller', 12, 'Male', 'USA'),
('Liu Wei', 6, 'Male', 'China'),
('Fatima Al-Sayed', 29, 'Female', 'Saudi Arabia');

-- mosaic_variants
INSERT INTO mosaic_variants
    (sample_id, gene_name, mutation_type, vaf_percent, is_harmful) VALUES
(1, 'MTOR',   'SNV',   25.40, 'Yes'),
(5, 'MTOR',   'Indel', 42.70, 'Yes'),
(7, 'NF1',    'CNV',   55.30, 'Yes'),
(8, 'GNAQ',   'SNV',   37.60, 'Yes');

-- surveillance_log
INSERT INTO surveillance_log
    (patient_id, variant_id, check_date, vaf_recorded, vaf_change, alert) VALUES
(1, 1, '2022-06-10', 25.40,  0.00, 'No'),
(1, 1, '2022-12-10', 29.80,  4.40, 'No'),
(1, 1, '2023-06-10', 36.20,  6.40, 'Yes');
```

5.3 SQL Queries

Query 1 — Selecting all the tables

```
USE sgss_rd;
SELECT * FROM patients;
SELECT * FROM tissue_samples;
SELECT * FROM mosaic_variants;
SELECT * FROM disease_diagnosis;
SELECT * FROM surveillance_log;
```

patient_id	patient_name	age	gender	country
1	John Smith	28	Male	USA
2	Sarah Johnson	35	Female	USA
3	Michael Chen	42	Male	USA
4	Emily Davis	19	Female	USA
5	David Wilson	53	Male	USA
6	Lisa Brown	31	Female	USA
7	Robert Taylor	27	Male	USA
8	Jennifer Anderson	44	Female	USA

sample_id	patient_id	organ_name	sample_date	sample_method
1	1	Brain	2022-03-10	Biopsy
2	2	Blood	2021-07-15	Blood
3	3	Skin	2023-01-22	Biopsy
4	4	Liver	2022-11-05	Biopsy
5	5	Heart	2021-09-18	Biopsy
6	6	Kidney	2023-04-12	Biopsy
7	7	Brain	2022-06-20	Biopsy
8	8	Blood	2021-12-03	Blood

diagnosis_id	patient_id	disease_name	diagnosis_date	how_confirmed	years_to_diagnose
1	1	Focal Cortical Dysplasia	2022-04-01	Both	3
2	2	Schimmelpenning Syndrome	2021-08-20	Clinical	5
3	3	Sturge-Weber Syndrome	2023-02-10	Both	2
4	4	Proteus Syndrome	2022-12-15	Genetic	4
5	5	Beckwith-Wiedemann Syndrome	2021-10-25	Both	1
6	6	Oculoectodermal Syndrome	2023-05-20	Clinical	6
7	7	Hemimegalencephaly	2022-07-30	Both	2
8	8	McCune-Albright Syndrome	2022-01-15	Genetic	3

log_id	patient_id	variant_id	check_date	vaf_recorded	vaf_change	alert
151	1	1	2022-06-10	25.40	0.00	No
152	1	1	2022-12-10	29.80	4.40	No
153	1	1	2023-06-10	36.20	6.40	Yes
154	2	2	2021-10-15	12.80	0.00	No
155	2	2	2022-04-20	15.20	2.40	No
156	2	2	2022-11-15	18.90	3.70	Yes
157	3	3	2023-04-22	31.50	0.00	No
158	3	3	2023-10-10	34.80	3.30	No

variant_id	sample_id	gene_name	mutation_type	vaf_percent	is_harmful
1	1	MTOR	SNV	25.40	Yes
2	2	HRAS	SNV	12.80	Yes
3	3	BRAF	SNV	31.50	Yes
4	4	PIK3CA	SNV	18.20	Yes
5	5	AKT1	SNV	22.10	Yes
6	6	MAP2K1	SNV	15.90	Yes
7	7	KRAS	SNV	28.60	Yes
8	8	NRAS	SNV	11.30	Yes

Query 2 — Selecting specific columns from surveillance_log

```
SELECT patient_id, check_date, vaf_recorded, alert
FROM surveillance_log;
```

patient_id	check_date	vaf_recorded	alert
1	2022-06-10	25.40	No
1	2022-12-10	29.80	No
1	2023-06-10	36.20	Yes
2	2021-10-15	12.80	No
2	2022-04-20	15.20	No
2	2022-11-15	18.90	Yes
3	2023-04-22	31.50	No
3	2023-10-10	34.80	No

Query 3 — Using SELECT DISTINCT statement

```
SELECT DISTINCT organ_name
FROM tissue_samples;
SELECT DISTINCT gene_name
FROM mosaic_variants;
SELECT DISTINCT mutation_type
FROM mosaic_variants;
SELECT DISTINCT country
FROM patients;
```

	organ_name
▶	Brain
	Blood
	Skin
	Liver
	Heart
	Kidney
	Muscle
	Eye

	gene_name
▶	MTOR
	HRAS
	BRAF
	PIK3CA
	AKT1
	MAP2K1
	KRAS
	NRAS

	mutation_type
▶	SNV
	DEL

	country
▶	USA
	France
	Germany
	Pakistan
	India

Query 4— Using SELECT DISTINCT statement for multiple columns

```
SELECT DISTINCT organ_name, sample_method
FROM tissue_samples
SELECT DISTINCT gene_name, mutation_type
FROM mosaic_variants;
```

	gene_name	mutation_type
▶	MTOR	SNV
	HRAS	SNV
	BRAF	SNV
	PIK3CA	SNV
	AKT1	SNV
	MAP2K1	SNV
	KRAS	SNV
	NRAS	SNV

	organ_name	sample_method
▶	Brain	Biopsy
	Blood	Blood
	Skin	Biopsy
	Liver	Biopsy
	Heart	Biopsy
	Kidney	Biopsy
	Muscle	Biopsy
	Eye	Biopsy

Query 5 — Using SELECT with WHERE clause

```
--Find all tissue samples collected from the Brain
SELECT * FROM tissue_samples
WHERE organ_name = 'Brain';
```

	sample_id	patient_id	organ_name	sample_date	sample_method
▶	1	1	Brain	2022-03-10	Biopsy
	7	7	Brain	2022-06-20	Biopsy
	11	11	Brain	2021-05-16	Biopsy
	18	18	Brain	2021-04-28	Biopsy
	25	25	Brain	2023-04-22	Biopsy
	31	31	Brain	2023-02-27	Biopsy
	37	37	Brain	2023-05-30	Biopsy
	45	45	Brain	2021-12-19	Biopsy
	NULL	NULL	NULL	NULL	NULL

```
-- Find surveillance entries where alert was triggered
SELECT * FROM surveillance_log
WHERE alert = 'Yes';
```

log_id	patient_id	variant_id	check_date	vaf_recorded	vaf_change	alert
153	1	1	2023-06-10	36.20	6.40	Yes
156	2	2	2022-11-15	18.90	3.70	Yes
159	3	3	2024-04-05	41.20	6.40	Yes
162	4	4	2024-02-20	24.50	4.40	Yes
165	5	5	2023-01-15	28.90	3.50	Yes
168	6	6	2024-07-10	22.10	3.80	Yes
171	7	7	2023-09-10	37.80	6.60	Yes
174	8	8	2023-03-20	17.40	3.60	Yes

Query 6 — Using SELECT with ORDER BY clause

```
-- List all variants ordered by VAF highest to lowest (DESC)
SELECT gene_name, vaf_percent, organ_name
FROM mosaic_variants
JOIN tissue_samples ON mosaic_variants.sample_id = tissue_samples.sample_id
ORDER BY vaf_percent DESC;
-- List all variants ordered by VAF highest to lowest (ASC)
SELECT gene_name, vaf_percent, organ_name
FROM mosaic_variants
JOIN tissue_samples ON mosaic_variants.sample_id = tissue_samples.sample_id
ORDER BY vaf_percent ASC;
```

gene_name	vaf_percent	organ_name
GNAQ	35.40	Eye
TP53	34.70	Skin
BRAF	33.60	Brain
GNAQ	33.20	Eye
BRAF	32.50	Skin
GNAS	32.10	Liver
NF1	31.80	Blood
BRAF	31.50	Skin

gene_name	vaf_percent	organ_name
NRAS	11.30	Blood
HRAS	12.80	Blood
GNAS	12.80	Skin
NF1	13.40	Kidney
NRAS	13.70	Heart
PTEN	14.50	Blood
PTEN	14.70	Heart
TSC1	15.30	Blood

Query 7 — Using Arithmetic Operators

```
-- Calculate how many healthy cells remain (100 - VAF%)
SELECT gene_name,
       vaf_percent AS mutated_cells_percent,
       100 - vaf_percent AS healthy_cells_percent
FROM mosaic_variants;

-- Calculate difference between vaf_recorded and vaf_change
SELECT log_id,
       vaf_recorded,
       vaf_change,
       vaf_recorded - vaf_change AS previous_vaf
FROM surveillance_log;
```

gene_name	mutated_cells_percent	healthy_cells_percent
MTOR	25.40	74.60
HRAS	12.80	87.20
BRAF	31.50	68.50
PIK3CA	18.20	81.80
AKT1	22.10	77.90
MAP2K1	15.90	84.10
KRAS	28.60	71.40
NRAS	11.30	88.70

log_id	vaf_recorded	vaf_change	previous_vaf
151	25.40	0.00	25.40
152	29.80	4.40	25.40
153	36.20	6.40	29.80
154	12.80	0.00	12.80
155	15.20	2.40	12.80
156	18.90	3.70	15.20
157	31.50	0.00	31.50
158	34.80	3.30	31.50

Query 8 — Using Operator Precedence

```
-- Calculate a risk score: (vaf_percent * 2) + years_to_diagnose
SELECT p.patient_name,
       m.vaf_percent,
       d.years_to_diagnose,
       (m.vaf_percent * 2) + d.years_to_diagnose AS risk_score
FROM patients p
JOIN tissue_samples t    ON p.patient_id = t.patient_id
JOIN mosaic_variants m   ON t.sample_id  = m.sample_id
JOIN disease_diagnosis d ON p.patient_id = d.patient_id;
```

patient_name	vaf_percent	years_to_diagnose	risk_score
John Smith	25.40	3	53.80
Sarah Johnson	12.80	5	30.60
Michael Chen	31.50	2	65.00
Emily Davis	18.20	4	40.40
David Wilson	22.10	1	45.20
Lisa Brown	15.90	6	37.80
Robert Taylor	28.60	2	59.20
Jennifer Anderson	11.30	3	25.60

Query 9 — Using Relational Operators

```
SELECT * FROM mosaic_variants
WHERE vaf_percent > 32;
```

variant_id	sample_id	gene_name	mutation_type	vaf_percent	is_harmful
9	9	TP53	SNV	34.70	Yes
13	13	GNAQ	SNV	33.20	Yes
20	20	BRAF	SNV	32.50	Yes
28	28	GNAQ	SNV	35.40	Yes
37	37	BRAF	SNV	33.60	Yes
44	44	GNAS	SNV	32.10	Yes
NULL	NULL	NULL	NULL	NULL	NULL

Query 10 — Using Logical Operators

```sql
-- Find harmful variants with high VAF or from blood
SELECT m.gene_name, m.vaf_percent, m.is_harmful, t.organ_name
FROM mosaic_variants m
JOIN tissue_samples t ON m.sample_id = t.sample_id
WHERE m.is_harmful = 'Yes'
  AND (m.vaf_percent > 30 OR t.organ_name = 'Blood');
```

gene_name	vaf_percent	is_harmful	organ_name
HRAS	12.80	Yes	Blood
BRAF	31.50	Yes	Skin
NRAS	11.30	Yes	Blood
TP53	34.70	Yes	Skin
PTEN	14.50	Yes	Blood
GNAQ	33.20	Yes	Eye
HRAS	16.20	Yes	Blood
BRAF	32.50	Yes	Skin
NRAS	[illegible]	Yes	Brain

Query 11 — Using LIKE Operator

```sql
-- Find disease names that contain the word 'Syndrome'
SELECT disease_name, years_to_diagnose
FROM disease_diagnosis
WHERE disease_name LIKE '%Syndrome%';
```

disease_name	years_to_diagnose
Schimmelpenning Syndrome	5
Sturge-Weber Syndrome	2
Proteus Syndrome	4
Beckwith-Wiedemann Syndrome	1
Oculoectodermal Syndrome	6
McCune-Albright Syndrome	3
CLAPO Syndrome	3
Epidermal Nevus Syndrome	7

Query 12 — Using LIKE Operator

```sql
-- Find all 5 functions together on mosaic_variants
SELECT COUNT(*)                    AS total_variants,
       ROUND(AVG(vaf_percent), 2)  AS avg_vaf,
       MAX(vaf_percent)            AS max_vaf,
       MIN(vaf_percent)            AS min_vaf,
       ROUND(SUM(vaf_percent), 2)  AS sum_vaf
FROM mosaic_variants;
```

total_variants	avg_vaf	max_vaf	min_vaf	sum_vaf
50	22.95	35.40	11.30	1147.50

6. Data Flow Diagrams

6.1 DFD Level 0 — Context:

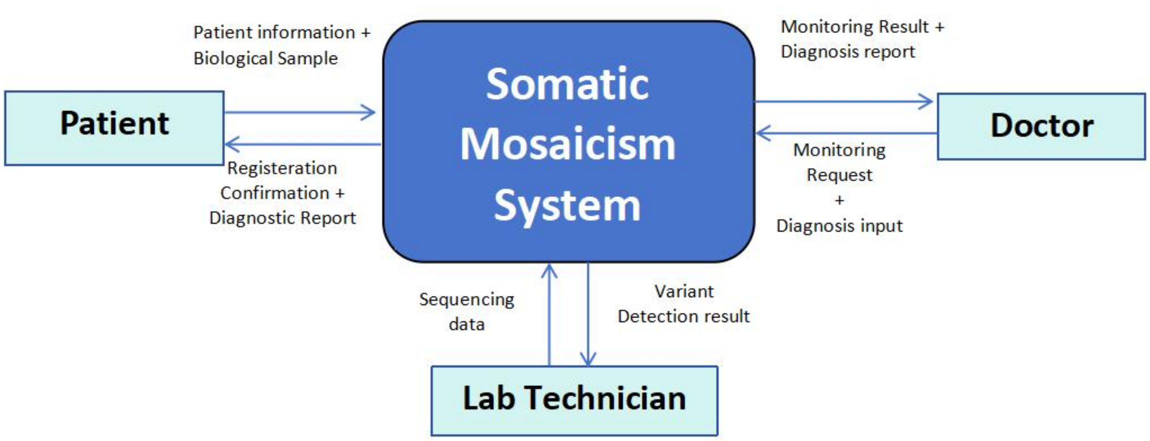
The Level 0 context diagram presents SGSS-RD as a single system process exposing only its external boundaries. Three categories of external entity supply data to the system: doctors provide patient demographic and medical history records; laboratory technicians upload biological sample metadata and sequencing results; patients provide informed consent and symptom histories. Three categories of external entity receive data from the system: specialist physicians receive diagnosis and variant reports; researchers receive anonymised variant datasets; health authorities receive epidemiological statistics; and the alert subsystem receives VAF threshold notifications for urgent clinical review.

6.2 DFD Level 1 — Detailed Processes:

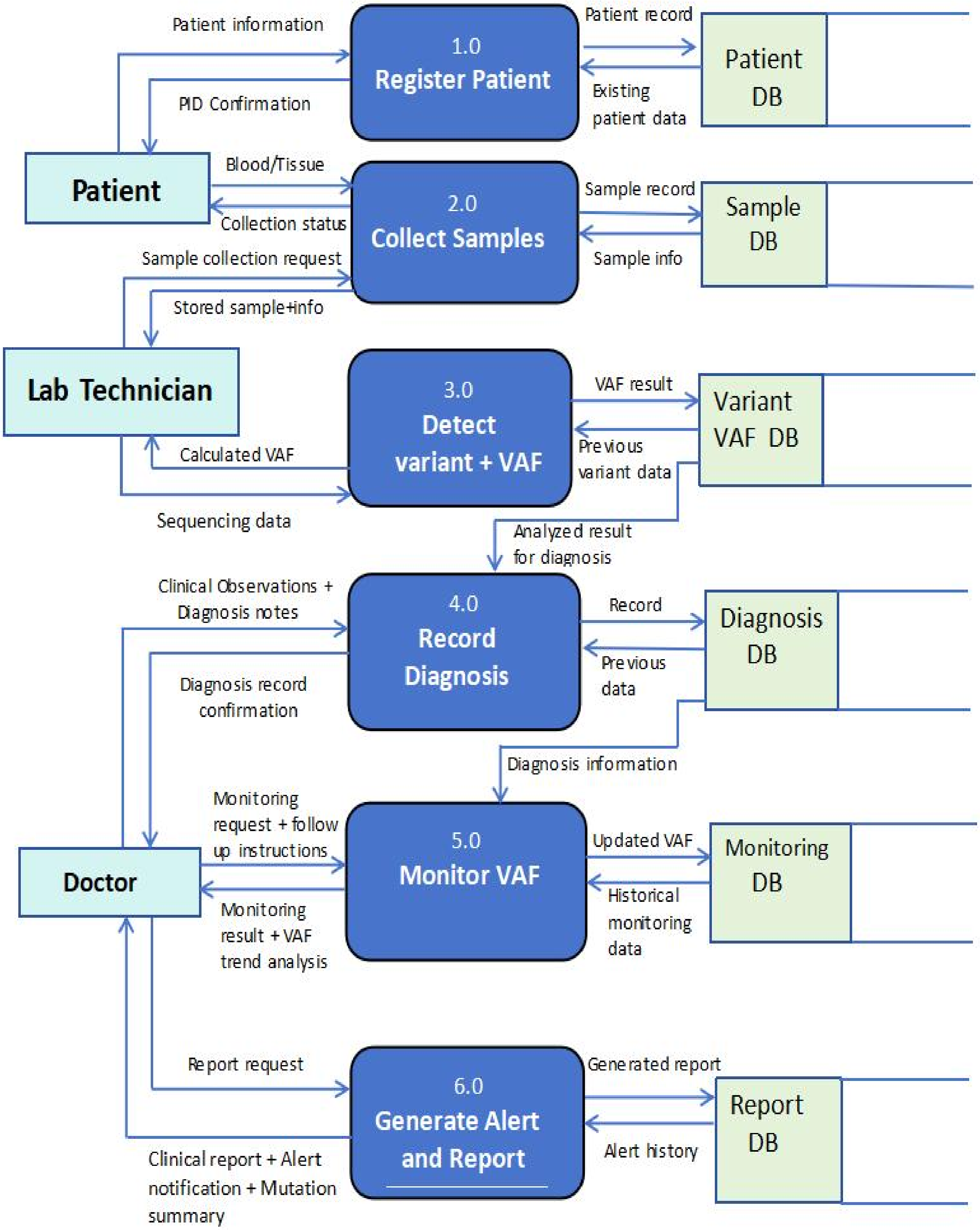
The Level 1 diagram decomposes the system into six distinct sub-processes, each corresponding to a major operational function. Five data stores are shown with their read and write flows.

- **P1 — Register Patient:** Accepts demographic and consent data from the clinician and patient, validates completeness, and writes a new record to the patients data store (D1).
- **P2 — Collect Sample:** Accepts biological sample metadata from the laboratory technician, reads the patient record from D1 to verify the patient identifier, and writes a new record to tissue_samples (D2).
- **P3 — Detect Variant and VAF:** Reads the sample record from D2, stores the sequencing-derived variant call and VAF percentage in mosaic_variants (D3), and exports anonymized data to researchers.
- **P4 — Record Diagnosis:** Accepts formal diagnosis information from the specialist physician, reads supporting data from D3 and D1, and writes the completed record to disease_diagnosis (D4).
- **P5 — Monitor VAF:** Reads variant and patient identifiers from D3 and D1, computes the VAF change relative to the previous log entry, and writes the time-stamped measurement to surveillance_log (D5).
- **P6 — Generate Alert and Report:** Reads aggregated data from D5, emits real-time alerts when VAF change exceeds 5 percent, and produces epidemiological trend reports for health authorities.
-

DFD Level 0 Diagram:

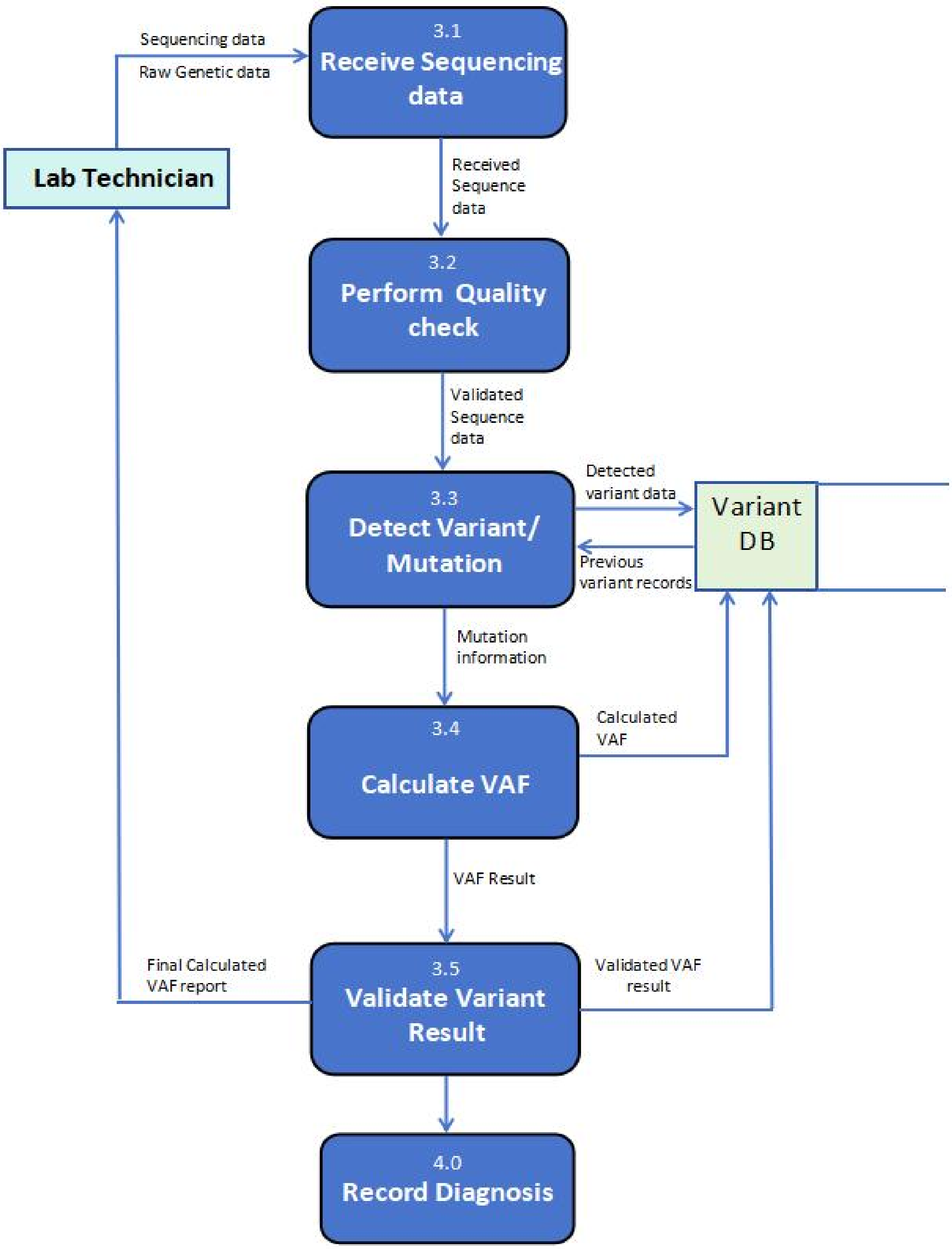


DFD Level 1 Diagram:



DFD Level 2 Diagram:

Decomposing Process 3.0(Detect Variant + VAF):



7. Results and Discussion

7.1 Schema Validation

The SGSS-RD schema was implemented and tested in MySQL 8.0. All five CREATE TABLE statements executed without error. Foreign key constraints were verified by attempting to insert child records with non-existent parent identifiers; MySQL correctly rejected all such attempts with constraint violation errors. Fifty rows of representative data were successfully inserted across all five tables, producing a consistent and fully traversable relational dataset.

7.2 Surveillance Demonstration

The surveillance functionality of SGSS-RD was demonstrated using Patient 1 (Ahmed Raza), who carries an MTOR SNV identified in a brain biopsy. Three surveillance log entries spanning twelve months show VAF progression from 25.40 percent to 29.80 percent to 36.20 percent. The third entry triggered an alert with a vaf_change value of 6.40, exceeding the 5.00 percent threshold. This time series illustrates the system's ability to detect clinically significant mutation spread that would be entirely invisible to a static variant registry.

7.3 Limitations and Future Work

The current implementation stores patient names as plaintext. Production deployment would require anonymisation tokens, role-based access control, and regulatory compliance infrastructure. The 5 percent VAF alert threshold is hardcoded in application logic rather than stored as a configurable database parameter; a future configuration table would allow clinical teams to adjust thresholds per gene or disease. Integration with OMIM, Orphanet, HPO, and ClinVar identifiers would further enhance interoperability with global genomic databases and enable federated queries across institutions.

8. Conclusion

The Structured Genomic Surveillance System for Rare Disorders demonstrates that a carefully normalised relational database schema can meaningfully advance the clinical management of somatic mosaicism. By organising data across five interrelated tables,The schema's most significant contribution is its surveillance architecture. The surveillance_log table records not simply what mutation a patient has, but how that mutation behaves over time. The schema is fully normalised to Third Normal Form, all foreign key relationships are enforced at the database level, fifty rows of representative data populate the system, and twelve SQL queries demonstrate analytical capabilities. The accompanying Entity Relationship Diagram and two-level Data Flow Diagrams provide complete documentation of the system's structure and process flows.Most importantly, SGSS-RD demonstrates that thoughtful database design grounded in a genuine clinical problem produces a system with real potential to reduce the diagnostic odyssey experienced by patients with rare mosaic genetic conditions.